

Cargo Calibration Protocol

Problem ID: kattisworkdir

You are an operations engineer aboard the interstellar freighter Orion's Reach. The ship is preparing for a hyperspace jump, but before that can happen, its cargo loadout must be calibrated carefully. Each crate on the ship contains modules with varying mass values. Every crate has a specific set of components that could be loaded, each with a defined weight. The ship's core AI randomly selects one component from each crate to include in the final cargo configuration. A successful configuration is one where the total cargo weight falls within a safe operational range for the hyperspace jump. Too light or too heavy, and the jump could fail catastrophically. Your task is to calculate the probability that a random cargo configuration will result in a total weight within the safe bounds.



Input

- The first line contains three integers: n – the number of cargo crates ($1 \leq n \leq 100$) min and max – the minimum and maximum acceptable total weight for the jump ($1 \leq min = max \leq 1,000$)
- The following n lines describe each crate: First an integer s ($1 \leq s \leq 100$), the number of available components for that crate, followed by s integers, each representing the weight of one component option

Output

Print the probability (as a percentage, rounded to two decimal places) that a randomly selected configuration of components results in a total weight between min and max , inclusive.

Note that a margin of error of $\pm 1\%$ is acceptable.

Sample Input 1

```
3 3 6
4 1 2 3 4
5 1 2 3 4 5
6 1 2 3 4 5 6
```

Sample Output 1

```
16.66%
```

Sample Input 2

```
2 5 5
2 2 10
2 3 5
```

Sample Output 2

```
25.00%
```